

Stackbirds Spring Internship 2026

Scenario

A company receives an invoice via email (PDF or image). You are creating an agent that will, for every invoice:

1. Extract structured invoice data
(Vendor name, line items, quantities, unit price, total, tax, etc.)
2. Verify that the vendor exists in the company's Approved Vendor list (provided as an Excel file).
3. Compare invoice line items against contracted rates in the Excel file.
4. Decide whether to:
 - "Approve" automatically
 - "Flag" for human review

The system must prioritize safety over automation.

Inputs We Provide

You will receive:

- 3 sample invoice files
 - [One clean](#)
 - [One messy](#)
 - [One edge-case scenario](#)
- An [Excel file](#) containing:
 - Approved vendor names (with possible aliases)
 - Contracted rates by SKU/service
 - Allowed price variance threshold
 - Notes on tax/shipping policies

Assume this Excel file is the source of truth.

Required Output

Your system must produce:

1. Structured Decision Payload (JSON)

Example structure:

- status: APPROVED or FLAGGED
- vendor_match_confidence
- variance_detected (true/false)

2. Human-Readable Reconciliation Report

Must include:

- Line-by-line comparison table
- Invoice price vs contracted price
- Variance calculation
- Clear explanation of decision
- Reasonably professional looking UI with some interactions / animations (feel free to select colors and style)

3. Audit Trail

A log of:

- What was extracted
- What assumptions were made
- Where uncertainty occurred
- Why the final decision was reached

We care deeply about explainability.

Clarification Behavior (Important)

If your system encounters ambiguity (e.g., vendor name mismatch, OCR uncertainty, missing line item), it must either:

- Print 1–3 specific clarifying questions
or
- Flag for review with a clear explanation

Do not silently assume correctness.

Technical Constraints

You may use any tools or stack (Python, n8n, composio, LLM APIs, etc.).

You may parse the Excel file directly or access it via a tool interface.

We are not evaluating specific frameworks.

We are evaluating how you think.

Deliverables

Please submit:

- A short architecture explanation (max 1 page)
- Your implementation (github link and instructions)
- Example outputs for each sample invoice
- A short section: “Top 5 risks in this system and how I would mitigate them”

If your instinct is to automate aggressively without supervision, rethink it :)